Prithvi Chakravarthy

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Education

DePaul University Chicago, IL
Master of Science in Computer Science *Sept. 2025 – Present*

- Coursework: Distributed Systems, Applied Algorithms and Structures, DBMS, Real-Time System Architecture

BITS Pilani, WILP India
Postgraduate Certificate in AI/ML *Oct. 2023 – Oct. 2024*

- Coursework: Deep Learning, Neural Networks, Text Mining, Feature Engineering, Unsupervised Learning

VIT Bhopal India
Bachelor of Technology in Computer Science *Jun. 2018 – Jun. 2022*

- Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Computer Architecture, Soft Computing, Computer Networks, Theory of Computation & Compiler Design

Experience

Designare Solutions Pvt Ltd Mumbai, India
Full Stack Developer *March 2022 – August 2025*

- Developed and deployed full stack applications using React, Node.js, Python (FastAPI) and PostgreSQL, improving scalability and backend performance.
- Designed distributed REST APIs and optimized data pipelines, reducing query retrieval time by 25%.
- Managed production deployments on Google Cloud infrastructure, maintaining 99.9% uptime.
- Implemented secure user authentication, role-based access control (RBAC), and wired external data sources into cohesive end-user interfaces.

Projects

AI Journal OS | *Rust, Tauri, React, TypeScript, SQLite, Candle*

- Built a privacy-first, local-offline AI ecosystem using a dual-topology architecture, integrating a Tauri React frontend with a standalone Rust background worker.
- Engineered a local Retrieval-Augmented Generation (RAG) pipeline utilizing pure Rust vector math and SQLite BLOB structures, generating CPU-friendly embeddings with the all-MiniLM-L6-v2 model.
- Designed a privacy-safe AI orchestration pipeline that budgets token counts, redacts PII locally via Hugging Face models, and routes sanitized context to the Groq API before rehydrating tokens upon response.
- Implemented a zero-knowledge cloud synchronization engine with Neon PostgreSQL, ensuring all third-party credentials and personal data remain AES-256 encrypted at rest.

Image Generation Model | *Python, Pytorch*

- Implemented a Stable Diffusion inference pipeline from scratch in PyTorch — VAE encoder/decoder for latent space compression, U-Net with multi-head cross-attention for iterative denoising, and CLIP ViT text conditioning
- Mapped pre-trained SD v1.5 weights into custom architecture; built both text-to-image and image-to-image pipelines.

Similar Image Recommender | *Python, Tensorflow, Keras*

- Built a deep learning image similarity engine using TensorFlow autoencoders and CNN-based feature embeddings to quantify visual similarity across large image sets.

Image Description Generator | *Python, Tensorflow, Jupyter Notebook*

- Developed a multi-modal image captioning model in TensorFlow generating natural language descriptions from visual input — combining CNN visual encoders with sequence generation.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, Rust, Kotlin, C#
ML Frameworks & Libraries: PyTorch, TensorFlow, Keras, OpenCV
AI/ML: Stable Diffusion, VAE, U-Net, CNNs, CLIP, Transformers, Autoencoders, LLMs, RAG, Embeddings, Multi-modal Models
Frameworks & Tools: React, Node.js, FastAPI, PostgreSQL, SQLite, Docker, Google Cloud, Git